File

Name: P3-K03-premachining
Description: KISSsoft example
Changed by: AAA on: 30.10.2026 at: 19:33:56

## 1 Overview

| | | | |
|---|---|---|---|
| Calculation method | | ISO 6336:2019 | |
| Drawing or article number: | | | |
| Gear 1: | | 0.000.0 | |
| Gear 2: | | 0.000.0 | |
| | | ------- Gear 1 | ---------- Gear 2 ------- |
| Power (kW) | [P] | 8.671 | |
| Speed (1/min) | [n] | 1200.0 | 320.0 |
| Number of load cycles (in mio.) | [NL] | 72.000 | 19.200 |
| Torque (Nm) | [T] | 69.0 | 258.8 |
| Application factor | [KA] | 1.10 | |
| Required service life (h) | [H] | 1000.00 | |
| Gear driving (+) / driven (-) | | + | - |
| Working flank Gear 1: | | Right flank | |
| Gear 1 direction of rotation: | | Clockwise | |

## 2 Tooth geometry

| | | | |
|---|---|---|---|
| Geometry calculation according to | | ISO 21771:2007 (withdrawn) | |
| | | ------- Gear 1 | ---------- Gear 2 ------- |
| Center distance (mm) | [a] | 72.500 | |
| Center distance tolerance | | ISO 286:2010 Measure js7 | |
| Normal module (mm) | [mn] | 2.5000 | |
| Transverse module (mm) | [mt] | 2.5000 | |
| Normal diametral pitch (1/in) | [Pnd] | 10.16000 | |
| Transverse diametral pitch (1/in) | [Ptd] | 10.16000 | |
| Normal pressure angle (°) | [αn] | 20.0000 | |
| Helix angle at reference circle (°) | [β] | 0.0000 | |
| Number of teeth | [z] | 12 | 45 |
| Facewidth (mm) | [b] | 50.00 | 48.00 |
| Hand of gear | | Spur gear | |
| Accuracy grade | [Q-ISO 1328:2013] | A6 | A7 |
| Inner diameter (mm) | [di] | 12.00 | 84.06 |
| Shaft diameter (mm) | [ds] | 12.00 | 84.06 |
| Inner diameter of gear rim (mm) | [dbi] | 0.00 | 0.00 |
| | | ------- Gear 1 | ---------- Gear 2 ------- |

## 3 Materials

**Gear 1**
18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

**Gear 2**
34CrAlMo5-10, Nitriding steel, gas-nitrided, ISO 6336-5 Figure 13a/14a (MQ)

| | | ------- Gear 1 ---------- | Gear 2 ------- |
|---|---|---|---|
| Surface hardness | | HRC 61 | HV 950 |
| Material treatment according to ISO 6336:2006 Normal, Life factors $Z_{NT}$ and $Y_{NT}$ ≥ 0.85 | | | |
| Infinite life strength for tooth root stress (N/mm²) | [σFlim] | 425.00 | 420.00 |
| Infinite life strength for Hertzian pressure (N/mm²) | [σHlim] | 1500.00 | 1250.00 |
| Young's modulus (N/mm²) | [E] | 206000 | 206000 |
| Poisson's ratio | [v] | 0.300 | 0.300 |
| Tensile strength (N/mm²) | [σB] | 1200.00 | 800.00 |
| Yield point (N/mm²) | [σS] | 850.00 | 600.00 |

### 3.1 Gear roughness

| | | ------- Gear 1 ---------- | Gear 2 ------- |
|---|---|---|---|
| Arithmetic mean roughness value $R_a$, flank (µm) | [RAH] | 0.60 | 2.00 |
| Arithmetic mean roughness value $R_a$, root (µm) | [RAF] | 3.00 | 2.00 |
| Mean peak-to-valley roughness $R_z$, flank (µm) | [RZH] | 4.80 | 14.00 |
| Mean peak-to-valley roughness $R_z$, root (µm) | [RZF] | 20.00 | 14.00 |

### 3.2 Lubrication

| | | |
|---|---|---|
| Lubrication type | Oil bath lubrication | |
| Type of oil | ISO-VG 220 | |
| Lubricant base | Mineral-oil base | |
| Lubricant factor | [XL] | 1.000 |
| Oil nominal kinematic viscosity at 40°C (mm²/s) | [v40] | 220.00 |
| Oil nominal kinematic viscosity at 100°C (mm²/s) | [v100] | 17.50 |
| Specific density at 15°C (kg/dm³) | [ρ] | 0.895 |
| Oil temperature (°C) | [TS] | 70.000 |

## 4 Geometry

### 4.1 Reference profiles

#### 4.1.1 Information on pre-machining

**Tool reference profile from Gear 1**

| | | |
|---|---|---|
| Cutter label, own input | | |
| Min. finishing stock (mm) | [qmin] | 0.110 |
| Addendum coefficient | [haP0*] | 1.386 |
| Dedendum coefficient | [hfP0*] | 1.200 |
| Tip radius factor | [ρaP0*] | 0.200 |
| | [ρaP0max*] | 0.401 |
| Root radius factor | [ρfP0*] | 0.001 |
| Protuberance height coefficient | [hprP0*] | 0.000 |
| Protuberance angle (°) | [αprP0] | 0.000 |
| Root form height coefficient | [hFfP0*] | 0.000 |
| Ramp angle (°) | [αKP0] | 0.000 |
| Tooth thickness coefficient reference line | [sP0*] | 1.571 |
| | not topping | |
| Addendum coefficient reference profile Gear | [haP*] | 1.000 |
| Manufacturing center distance (mm) | [a0] | 16.609 |
| (mm) | [a0.e/i] | 16.535 /16.494 |
| Smallest radius of curvature, root rounding (mm) | [ρmin.e/i] | 0.624 / 0.631 |

**Tool reference profile from Gear 2**

| | | |
|---|---|---|
| Cutter label, own input | | |
| Min. finishing stock (mm) | [qmin] | 0.100 |
| Protuberance (mm) | [pr0] | 0.150 |
| Remaining undercut (mm) | [spr0 = pr0-q] | 0.050 |

| | | |
|---|---|---|
| Addendum coefficient | [haP0*] | 1.300 |
| Dedendum coefficient | [hfP0*] | 1.200 |
| Tip radius factor | [ρaP0*] | 0.250 |
| | [ρaP0max*] | 0.498 |
| Root radius factor | [ρfP0*] | 0.001 |
| Protuberance height coefficient | [hprP0*] | 0.603 |
| Protuberance angle (°) | [αprP0] | 12.000 |
| Root form height coefficient | [hFfP0*] | 0.000 |
| Ramp angle (°) | [αKP0] | 0.000 |
| Tooth thickness coefficient reference line | [sP0*] | 1.571 |
| | not topping | |
| Addendum coefficient reference profile Gear | [haP*] | 1.000 |
| Manufacturing center distance (mm) | [a0] | 56.679 |
| (mm) | [a0.e/i] | 56.583 /56.528 |
| Smallest radius of curvature, root rounding (mm) | [ρmin.e/i] | 0.715 / 0.719 |

**4.1.2 Reference profile of the final tooth form**

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Dedendum reference profile | [hfP*] | 1.241 | 1.161 |
| Tooth root radius reference profile | [ρfP*] | 0.200 | 0.250 |
| Addendum reference profile | [haP*] | 1.000 | 1.000 |
| Protuberance height coefficient | [hprP*] | 0.000 | 0.269 |
| Protuberance angle (°) | [αprP] | 0.000 | 12.000 |
| Tip form height coefficient | [hFaP*] | 0.000 | 0.000 |
| Ramp angle (°) | [αKP] | 0.000 | 0.000 |
| | | | |
| Data for final machining: | | | |
| Depth of immersion | [hgrind*] | 1.071 | 0.800 |
| Radius at cutter head | [rgrind*] | 0.100 | 0.100 |
| Grinding only flank (0), flank & root (1) | | 0 | 0 |
| Generation grinding (0), form grinding (1) | | 0 | 0 |
| | | | |
| Type of profile modification: | none (only running-in) | | |
| Tip relief by running in (µm) | [Ca L/R] | 2.0 / 2.0 | 3.2 / 3.2 |

**4.2 Basic data**

| | | |
|---|---|---|
| Overall transmission ratio | [itot] | -3.750 |
| Gear ratio | [u] | 3.750 |
| Transverse module (mm) | [mt] | 2.500 |
| Transverse pressure angle (°) | [αt] | 20.000 |
| Working pressure angle (°) | [αwt] | 22.558 |
| (°) | [αwt.e/i] | 22.587 /22.530 |
| Working pressure angle at normal section (°) | [αwn] | 22.558 |
| Base helix angle (°) | [βb] | 0.000 |
| Helix angle at operating pitch circle (°) | [βw] | 0.000 |
| Reference center distance (mm) | [ad] | 71.250 |
| Sum of profile shift coefficients | [Σxi] | 0.5313 |

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Profile shift coefficient | | | |
| Information on pre-machining | [xV] | 0.6437 | 0.1716 |
| Information on final machining | [x] | 0.4986 | 0.0327 |
| | | | |
| Generating profile shift coefficient | | | |
| Information on pre-machining | [xEV.e/i] | 0.6140/ 0.5975 | 0.1332/ 0.1112 |
| Information on final machining | [xE.e/i] | 0.4689/ 0.4524 | -0.0058/-0.0278 |
| Virtual gear no. of teeth | [zn] | 12.000 | 45.000 |
| Involute length (mm) | [l_dFa-l_dFf] | 5.346 | 4.388 |

## 4.3 Diameters and their allowances

| | | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|---|
| Reference diameter (mm) | | [d] | 30.000 | 112.500 |
| Base diameter (mm) | | [db] | 28.191 | 105.715 |
| Tip alteration (mm) | | [k*mn] | 0.000 | -0.125 |
| Tip diameter (mm) | | [da] | 37.493 | 117.413 |
| (mm) | | [da.e/i] | 37.493 /37.483 | 117.413 /117.403 |
| Tip diameter allowances (mm) | | [Ada.e/i] | 0.000 / -0.010 | 0.000 / -0.010 |
| Tip form diameter (mm) | | [dFa] | 37.493 | 117.413 |
| (mm) | | [dFa.e/i] | 37.493 /37.483 | 117.413 /117.403 |
| Root form diameter, pre-machining (mm) | | [dFa_p] | 37.493 | 117.413 |
| | (mm) | [dFa_p.e/i] | 37.493 /37.483 | 117.413 /117.403 |
| Active tip diameter (mm) | | [dNa] | 37.493 | 117.413 |
| (mm) | | [dNa.e/i] | 37.493 /37.483 | 117.413 /117.403 |
| V-Circle diameter (mm) | | [dv] | 32.493 | 112.663 |
| (mm) | | [dv.e/i] | 32.344 /32.262 | 112.471 /112.361 |
| Operating pitch diameter (mm) | | [dw] | 30.526 | 114.474 |
| (mm) | | [dw.e/i] | 30.533 /30.520 | 114.497 /114.450 |
| Root diameter (mm) | | [df] | 26.290 | 106.858 |
| (mm) | | [df.e/i] | 26.142 /26.059 | 106.666 /106.556 |
| (calculated with pre-machining tool) | | | | |
| Active root diameter (mm) | | [dNf] | 28.553 | 110.141 |
| (mm) | | [dNf.e/i] | 28.569 /28.541 | 110.167 /110.119 |
| Root form diameter (mm) | | [dFf] | 28.335 | 109.226 |
| (mm) | | [dFf.e/i] | 28.294 /28.275 | 109.274 /109.052 |
| (calculated with final machining tool) | | | | |
| Root form diameter, pre-machining (mm) | | [dFf_p] | 28.222 | 110.302 |
| | (mm) | [dFf_p.e/i] | 28.205 /28.199 | 110.180 /110.115 |

## 4.4 Tip clearances and tooth heights

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Theoretical tip clearance (mm) | [c] | 0.325 | 0.648 |
| Effective tip clearance (mm) | [c.e/i] | 0.496 / 0.406 | 0.784 / 0.707 |
| Reserve (dNf-dFf)/2 (mm) | [cF.e/i] | 0.147 / 0.123 | 0.557 / 0.422 |
| Undercut taken into account | | | |
| Addendum, $m_n(h_{aP}*+x+k)$ (mm) | [ha] | 3.746 | 2.457 |
| (mm) | [ha.e/i] | 3.746 / 3.741 | 2.457 / 2.452 |
| Dedendum, $m_n(h_{fP}*-x)$ (mm) | [hf] | 1.855 | 2.821 |
| (mm) | [hf.e/i] | 1.929 / 1.970 | 2.917 / 2.972 |
| Tooth height (mm) | [h] | 5.601 | 5.278 |

## 4.5 Roll angle

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Roll angle at dFa (°) | [ξFa.e/i] | 50.238 /50.207 | 27.690 /27.677 |
| Roll angle to dNf (°) | [ξNf.e/i] | 9.423 / 9.058 | 16.801 /16.708 |
| Roll angle at dFf (°) | [ξFf.e/i] | 4.918 / 4.428 | 14.991 /14.509 |

**4.6 Tooth thickness and pitch**

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Tooth thickness on reference circle, arc, in module | [sn*] | 1.9337 | 1.5946 |
| Normal tooth thickness at tip circle (mm) | [san] | 0.716 | 2.025 |
| (mm) | [san.e/i] | 0.657 / 0.611 | 1.956 / 1.910 |
| Normal tooth thickness at tip form circle (mm) | [sFan] | 0.716 | 2.025 |
| (mm) | [sFan.e/i] | 0.657 / 0.611 | 1.956 / 1.910 |
| Normal space width at root circle (mm) | [efn] | 0.000 | 2.194 |
| (mm) | [efn.e/i] | 0.000 / 0.000 | 2.229 / 2.250 |
| Pitch on reference circle (mm) | [pt] | 7.854 | 7.854 |
| Base pitch (mm) | [pbt] | 7.380 | 7.380 |
| Transverse pitch on contact-path (mm) | [pet] | 7.380 | 7.380 |

**4.7 Sliding**

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Max. sliding velocity at tip (m/s) | [vga] | 1.035 | 0.571 |
| Specific sliding at the tip | [ζa] | 0.667 | 0.667 |
| Specific sliding at the root | [ζf] | -2.004 | -1.999 |
| Mean specific sliding | [ζm] | 0.667 | |
| Sliding factor on tip | [Kga] | 0.540 | 0.298 |
| Sliding factor on root | [Kgf] | -0.298 | -0.540 |

**4.8 Contact ratios**

| | | ---------- Pair ---------- |
|---|---|---|
| Minimal length of contact line (mm) | [Lmin] | 48.000 |
| Transverse contact ratio | [εα] | 1.367 |
| | [εα.e/m/i] | 1.373 / 1.366/ 1.359 |
| Overlap ratio | [εβ] | 0.000 |
| Total contact ratio | [εγ] | 1.367 |
| | [εγ.e/m/i] | 1.373 / 1.366/ 1.359 |
| Length of path of contact (mm) | [gα] | 10.092 |
| (mm) | [gα.e/m/i] | 10.131 /10.082/10.033 |
| Approach path of contact (mm) | [gf] | 3.588 |
| (mm) | [gf.e/m/i] | 3.619 / 3.582/ 3.545 |
| Recess path of contact (mm) | [ga] | 6.504 |
| (mm) | [ga.e/m/i] | 6.512 / 6.500/ 6.488 |

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Addendum contact ratio | [ε] | 0.881 | 0.486 |
| | [ε.e/i] | 0.882 / 0.879 | 0.490 / 0.480 |
| Length T1-A and T2-A (mm) | [T1A,T2A] | 2.268 | 25.545 |
| (mm) | [.e/i] | 2.228 / 2.318 | 25.545 / 25.533 |
| Length T1-B and T2-B (mm) | [T1B,T2B] | 4.979 | 22.834 |
| (mm) | [.e/i] | 4.979 / 4.971 | 22.795 / 22.880 |
| Length T1-C and T2-C (mm) | [T1C,T2C] | 5.855 | 21.957 |
| (mm) | [.e/i] | 5.847 / 5.863 | 21.926 / 21.988 |

| | | | |
|---|---|---|---|
| Length T1-D and T2-D (mm) | [T1D,T2D] | 9.648 | 18.165 |
| (mm) | [.e/i] | 9.609 / 9.698 | 18.165 / 18.153 |
| Length T1-E and T2-E (mm) | [T1E,T2E] | 12.359 | 15.453 |
| (mm) | [.e/i] | 12.359 / 12.352 | 15.414 / 15.500 |
| Length T1-T2 (mm) | [T1T2] | 27.812 | |
| (mm) | [.e/i] | 27.773 / 27.852 | |
| Diameter of single contact point B (mm) | [d-B] | 29.898 | 115.158 |
| (mm) | [d-B.e/i] | 29.898 / 29.893 | 115.127 /115.195 |
| Diameter of single contact point D (mm) | [d-D] | 34.162 | 111.784 |
| (mm) | [d-D.e/i] | 34.118 / 34.219 | 111.784 /111.776 |

# 5 General influence factors

## 5.1 Forces and circumferential speed

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Nominal circum. force at pitch circle (N) | [Ft] | 4600.00 | |
| Axial force (N) | [Fa] | 0.00 | 0.00 |
| Radial force (N) | [Fr] | 1674.26 | 1674.26 |
| Normal force (N) | [Fnorm] | 4895.22 | 4895.22 |
| Nominal circumferential force per mm (N/mm) | [w] | 95.83 | |
| Only as information: Forces at operating pitch circle: | | | |
| Nominal circumferential force (N) | [Ftw] | 4520.69 | |
| Axial force (N) | [Faw] | 0.00 | 0.00 |
| Radial force (N) | [Frw] | 1877.90 | |
| Circumferential speed reference circle (m/s) | [v] | 1.88 | |
| Circumferential speed operating pitch circle (m/s) | [v(dw)] | 1.92 | |

## 5.2 Contact stiffness

| | | |
|---|---|---|
| Running-in value (µm) | [yp] | 0.8 |
| Running-in value (µm) | [yf] | 1.0 |
| Correction factor | [CM] | 0.800 |
| Gear blank factor | [CR] | 1.000 |
| Basic rack factor | [CB] | 1.000 |
| Material coefficient | [E/Est] | 1.000 |
| Single stiffness (N/mm/µm) | [c'] | 13.556 |
| Theoretical single stiffness (N/mm/µm) | [c'th] | 16.952 |
| Contact stiffness (N/mm/µm) | [cγα] | 17.291 |
| Contact stiffness (N/mm/µm) | [cγβ] | 14.697 |
| Reduced mass (kg/mm) | [mRed] | 0.00347 |
| Resonance speed (min-1) | [nE1] | 56211 |
| Resonance ratio (-) | [N] | 0.021 |
| Subcritical range | | |

## 5.3 Calculation of K factors

| | | | |
|---|---|---|---|
| Running-in value (µm) | [yα] | 1.0 | |
| Bearing distance l of pinion shaft (mm) | [l] | 100.000 | |
| Distance s of pinion shaft (mm) | [s] | 10.000 | |
| Outside diameter of pinion shaft (mm) | [dsh] | 27.270 | |
| Load in accordance with Figure 13, ISO 6336-1:2006 0:a), 1:b), 2:c), 3:d), 4:e) | [-] | | 4 |
| Coefficient K' according to Figure 13, ISO 6336-1:2006 Without stiffening | [K'] | | -1.00 |
| Effective flank line deviation (µm) | [Fβy] | 5.95 | |

| | | |
|---|---|---|
| from deformation of shaft (µm) | [fsh·B1] | 7.81 |
| fsh (µm) | [fsh] | 7.81 |
| B1 | [B1] | 1.00 |
| fHβ5 (µm) | [fHβ5] | 7.00 |
| Tooth without flank line modification | | |
| Position of contact pattern: | favorable | |
| from production tolerances (µm) | [fmα*B2] | 16.92 |
| $B_2$=1.00 | | |
| Tooth trace deviation, theoretical (µm) | [Fβx] | 7.00 |
| Running-in value (µm) | [yβ] | 1.05 |

### 5.4 K factors

| | | |
|---|---|---|
| Dynamic factor | [Kv] | 1.025 |
| Face load factors | | |
| - Flank | [KHβ] | 1.405 |
| - Tooth root | [KFβ] | 1.353 |
| - Scuffing | [KBβ] | 1.405 |
| Transverse load factors | | |
| - Flank | [KHα] | 1.000 |
| - Tooth root | [KFα] | 1.000 |
| - Scuffing | [KBα] | 1.000 |
| Application factor | [KA] | 1.100 |
| Mesh load factor | [Kγ] | 1.000 |

## 6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

| | | ------- Gear 1 ---------- | ---------- Gear 2 ------- |
|---|---|---|---|
| Calculated with generating profile shift coefficient | [xE.i] | 0.5975 | -0.0278 |
| Tooth form factor | [YF] | 1.568 | 1.833 |
| Stress correction factor | [YS] | 2.264 | 1.974 |
| Load application angle (°) | [αFen] | 29.291 | 21.923 |
| Load distribution influence factor | [fε] | 1.000 | |
| Load application diameter (mm) | [$d_{en}$] | 34.162 | 115.158 |
| Bending moment arm (mm) | [hFe] | 3.176 | 3.388 |
| (-) | [hFe*] | 1.270 | 1.355 |
| Tooth root thickness at critical cross section (mm) | [sFn] | 5.310 | 5.232 |
| (-) | [sFn*] | 2.124 | 2.093 |
| Root fillet radius at critical cross section (mm) | [ρF] | 0.791 | 1.036 |
| (-) | [ρF*] | 0.316 | 0.414 |
| Tangent contact point (mm) | [x,y] | 2.655 /12.986 | |
| Tangent contact point (mm) | [x,y] | | 2.616 /53.590 |
| Diameter at critical cross section (mm) | [$d_{sFn}$] | 26.509 | 107.308 |
| Tangent at critical cross section (°) | [$\alpha_{sFn}$] | 30.000 | 30.000 |
| Notch parameter | [$q_s$] | 3.357 | 2.525 |
| Helix angle factor | [Yβ] | 1.000 | |
| Deep tooth factor | [YDT] | 1.000 | |
| Gear rim factor | [YB] | 1.00 | 1.00 |
| Effective facewidth (mm) | [beff] | 50.00 | 48.00 |
| Nominal stress at tooth root (N/mm²) | [σF0] | 130.65 | 138.70 |
| Tooth root stress (N/mm²) | [σF] | 199.32 | 211.59 |
| Permissible bending stress at root of Test-gear | | | |
| Notch sensitivity factor | [YδrelT] | 1.008 | 1.001 |

| | | | |
|---|---|---|---|
| Surface factor | [YRrelT] | 0.957 | 0.996 |
| Size factor, tooth root | [YX] | 1.000 | 1.000 |
| Life factor | [YNT] | 0.938 | 0.963 |
| $Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$ | | 0.905 | 0.960 |
| Alternating bending factor, mean stress influence coefficient | [YM] | 1.000 | 1.000 |
| Stress correction factor | [YST] | 2.00 | |
| $Y_{ST} \cdot \sigma_{Flim}$ (N/mm²) | [σFE] | 850.00 | 840.00 |
| Permissible tooth root stress σFG/SFmin (N/mm²) | [σFP] | 549.18 | 576.16 |
| Limit strength tooth root (N/mm²) | [σFG] | 768.85 | 806.62 |

### 6.1 Safety factors

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Required safety | [SFmin] | 1.40 | 1.40 |
| Safety for tooth root stress | [SF=σFG/σF] | 3.86 | 3.81 |
| Transmittable power (kW) | [kWRating] | 23.89 | 23.61 |

## 7 Calculation of flank strength (pitting)

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Zone factor | [ZH] | 2.335 | |
| Elasticity coefficient (√N/mm) | [ZE] | 189.812 | |
| Contact ratio factor | [Zε] | 0.937 | |
| Helix angle factor | [Zβ] | 1.000 | |
| Effective facewidth (mm) | [beff] | 48.00 | |
| Nominal contact stress (N/mm²) | [σH0] | 835.18 | |
| Contact stress at operating pitch circle (N/mm²) | [σHw] | 1051.16 | |
| Single tooth contact factor | [ZB,ZD] | 1.06 | 1.00 |
| Contact stress (N/mm²) | [σHB, σHD] | 1117.85 | 1051.16 |
| Lubricant coefficient for $N_L$ | [ZL] | 1.020 | 1.020 |
| Speed factor at $N_L$ | [ZV] | 0.963 | 0.963 |
| Roughness factor for $N_L$ | [ZR] | 0.894 | 0.894 |
| Material hardening factor for $N_L$ | [ZW] | 1.000 | 1.000 |
| Life factor | [ZNT] | 0.989 | 0.958 |
| Size factor (flank) | [ZX] | 1.000 | 1.000 |
| $Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$ | | 0.869 | 0.841 |
| Limited pitting is permissible: | No | | |
| Permissible contact stress, σHG/SHmin (N/mm²) | [σHP] | 1302.97 | 1051.64 |
| Pitting stress limit (N/mm²) | [σHG] | 1302.97 | 1051.64 |

### 7.1 Safety factors

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Required safety | [SHmin] | 1.00 | 1.00 |
| Safety factor for contact stress on operating pitch circle | [SHw] | 1.24 | 1.00 |
| Safety against pressure, σHG/σHBD Single contact | [SHBD] | 1.17 | 1.00 |
| Safety regarding transmittable torque | [SHBD²] | 1.36 | 1.00 |
| Transmittable power (kW) | [kWRating] | 11.78 | 8.68 |

## 8 Micropitting

Calculation method according to ISO/TS 6336-22:2018

Calculation has not been carried out, lubricant: Load stage micropitting test not known

## 9 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Thermal contact coefficient (N/mm/s^.5/K) | [BM] | 13.780 | 13.780 |
| Relevant tip relief (µm) | [Ca] | 2.00 | 3.20 |
| Optimal tip relief (µm) | [Ceff] | 7.78 | |
| $C_a$ taken as optimal in the calculation. 0=no, 1=yes | | 0 | 0 |

| | | ---------- Pair ---------- | |
|---|---|---|---|
| Helical load factor for scuffing | [KBγ] | 1.000 | |
| Lubrication coefficient for lubrication type | [XS] | 1.000 | |
| Scuffing test, load stage | [FZGtest] | FZG - Test A / 8.3 / 90 (ISO 14635 - 1), | 12 |
| Multiple meshing factor | [Xmp] | 1.000 | |
| Relative structural factor, scuffing | [XWrelT] | 1.000 | |
| Effective facewidth (mm) | [beff] | 48.000 | |
| Applicable circumferential force/facewidth (N/mm) | [wBt] | 151.808 | |
| $K_{B\gamma}$ = 1.000, $w_{Bt} \cdot K_{B\gamma}$ = 151.808 | | | |
| Angle factor | [Xαβ] | 1.015 | |
| $\varepsilon_1$ =0.881, $\varepsilon_2$ =0.486 | | | |

**9.1 Flash temperature-criteria**

| | | ---------- Pair ---------- | |
|---|---|---|---|
| Lubricant factor | [XL] | 0.830 | |
| Tooth mass temperature (°C) | [θMi] | 76.43 | |
| $\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$ | | | |
| Average flash temperature (Δ°C) | [θflm] | 13.68 | |
| Scuffing temperature (°C) | [θS] | 348.81 | |
| Γ coordinates (point of highest temperature) | [Γ] | 0.641 | |
| [Γ.A]= -0.613, [Γ.E]= 1.111 | | | |
| Maximum flash temperature (Δ°C) | [θflamax] | 26.05 | |
| Maximum contact temperature (°C) | [θBmax] | 102.48 | |
| Flash factor (°K·N^-.75·s^.5·m^-.5·mm) | [XM] | 50.058 | |
| Approach factor | [XJ] | 1.000 | |
| Load sharing factor | [XΓ] | 1.000 | |
| Dynamic viscosity (mPa*s) | [ηM] | 41.87 | (70.0 °C) |
| Coefficient of friction | [$\mu_m$] | 0.087 | |
| Required safety | [SBmin] | 2.000 | |
| Margin of safety for scuffing, flash temperature | [SB] | 8.584 | |

**9.2 Integral temperature-criteria**

| | | ---------- Pair ---------- | |
|---|---|---|---|
| Lubricant factor | [XL] | 1.000 | |
| Tooth mass temperature (°C) | [θMC] | 77.99 | |
| $\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$ | | | |
| Integral scuffing temperature (°C) | [θSint] | 360.78 | |
| Flash factor (°K·N^-.75·s^.5·m^-.5·mm) | [XM] | 50.058 | |
| Running-in factor, well run in | [XE] | 1.000 | |
| Contact ratio factor | [Xε] | 0.279 | |
| Dynamic viscosity (mPa*s) | [ηOil] | 41.87 | (70.0 °C) |
| Mean coefficient of friction | [$\mu_m$] | 0.111 | |
| Geometry factor | [XBE] | 0.444 | |
| Meshing factor | [XQ] | 1.000 | |
| Tip relief factor | [XCa] | 1.247 | |
| Mean flash temperature (Δ°C) | [θflaint] | 11.41 | |
| Integral tooth flank temperature (°C) | [θint] | 95.11 | |
| Required safety | [SSmin] | 1.800 | |
| Safety factor for scuffing (intg.-temp.) | [SSint] | 3.793 | |
| Safety factor for transmitted moment (int.-T.) | [SSL] | 11.580 | |

## 10 Measurements for tooth thickness

### 10.1 Tooth thickness tolerances and inputs

| | | ------- Gear 1 ---------- | Gear 2 ------- |
|---|---|---|---|
| **10.1.1 Information on pre-machining** | | | |
| Min. finishing stock (mm) | [qmin] | 0.110 | 0.100 |
| Tooth thickness tolerance pre-machining (mm) | [TsnV] | 0.030 | 0.040 |
| Additional measure for pre-machining (mm) | [ΔAs_p.e/i] | 0.294 / 0.234 | 0.293 / 0.213 |
| Tooth thickness allowance (normal section) (mm) | [As_p.e/i] | 0.210 / 0.180 | 0.183 / 0.143 |
| **10.1.2 Information on final machining** | | | |
| Tooth thickness tolerance | | DIN 3967 cd25 | DIN 3967 cd25 |
| Tooth thickness allowance (normal section) (mm) | [Asn.e/i] | -0.054 / -0.084 | -0.070 / -0.110 |

### 10.2 Base tangent lengths

| | | ------- Gear 1 ---------- | Gear 2 ------- |
|---|---|---|---|
| **10.2.1 Information on pre-machining** | | | |
| Number of teeth spanned | [k] | 3.000 | 6.000 |
| Base tangent length with allowance (mm) | [Wk.e/i] | 19.921 /19.893 | 42.395 /42.358 |
| (mm) | [ΔWk.e/i] | 0.197 / 0.169 | 0.172 / 0.134 |
| **10.2.2 Information on final machining** | | | |
| Number of teeth spanned | [k] | 3.000 | 6.000 |
| Base tangent length (no backlash) (mm) | [Wk] | 19.724 | 42.223 |
| Base tangent length with allowance (mm) | [Wk.e/i] | 19.673 /19.645 | 42.158 /42.120 |
| (mm) | [ΔWk.e/i] | -0.051 / -0.079 | -0.066 / -0.103 |
| Diameter of measuring circle (mm) | [dMWk.m] | 34.368 | 113.804 |

### 10.3 Measurement over balls and pins

| | | ------- Gear 1 ---------- | Gear 2 ------- |
|---|---|---|---|
| **10.3.1 Information on pre-machining** | | | |
| Effective diameter of ball/pin (mm) | [DMeff] | 6.000 | 4.250 |
| Diametral measurement over two balls (mm) | [MdK.e/i] | 42.497 /42.452 | 118.836 /118.739 |
| **10.3.2 Information on final machining** | | | |
| Theoretical diameter of ball/pin (mm) | [DM] | 5.672 | 4.248 |
| Effective diameter of ball/pin (mm) | [DMeff] | 6.000 | 4.250 |
| Radial single-ball measurement, no backlash (mm) | [MrK] | 21.092 | 59.229 |
| Radial single-ball measurement (mm) | [MrK.e/i] | 21.051 /21.029 | 59.141 /59.091 |
| Diameter of measuring circle (mm) | [dMMr.m] | 32.674 | 112.462 |
| Diametral measurement over two balls, no backlash (mm) | [MdK] | 42.184 | 118.388 |
| Diametral measurement over two balls (mm) | [MdK.e/i] | 42.103 /42.058 | 118.213 /118.113 |
| Diametral measurement over pins, no backlash (mm) | [MdR] | 42.184 | 118.388 |
| Measurement over pins according to DIN 3960 (mm) | [MdR.e/i] | 42.103 /42.058 | 118.213 /118.113 |
| Measurement over 3 pins, no backlash (mm) | [Md3R] | 0.000 | 118.318 |
| Measurement over 3 pins (mm) | [Md3R.e/i] | 0.000 / 0.000 | 118.144 /118.043 |

## 10.4 Tooth thickness

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| **10.4.1 Information on pre-machining** | | | |
| Medium tip diameter (mm) | [da.m] | 37.488 | 117.408 |
| Reference chordal height from da.m (mm) | [hac] | 3.955 | 2.492 |
| Tooth thickness at height hac, chord (mm) | [sc.e/i] | 5.020 / 4.992 | 4.168 / 4.129 |
| Tooth thickness on reference circle, arc (mm) | [sn.e/i] | 5.044 / 5.014 | 4.169 / 4.129 |
| **10.4.2 Information on final machining** | | | |
| Medium tip diameter (mm) | [da.m] | 37.488 | 117.408 |
| Reference chordal height from da.m (mm) | [hac] | 3.939 | 2.490 |
| Tooth thickness at height hac, chord (mm) | [sc] | 4.813 | 3.986 |
| (mm) | [sc.e/i] | 4.762 / 4.733 | 3.916 / 3.877 |
| Tooth thickness on reference circle, arc (mm) | [sn] | 4.834 | 3.986 |
| (mm) | [sn.e/i] | 4.780 / 4.750 | 3.916 / 3.876 |

## 10.5 Backlash

| | | ---------- Pair ---------- |
|---|---|---|
| Circumferential backlash from Aa (mm) | [jtw_Aa.e/i] | 0.012 / -0.012 |
| Radial backlash (mm) | [jrw.e/i] | 0.255 / 0.138 |
| Circumferential backlash (transverse section) (mm) | [jtw.e/i] | 0.210 / 0.114 |
| Normal backlash (mm) | [jnw.e/i] | 0.193 / 0.106 |
| Center distance allowances (mm) | [Aa.e/i] | 0.015 / -0.015 |
| Backlash free center distance (mm) | [aControl.e/i] | 72.347 / 72.260 |
| Backlash free center distance, allowances (mm) | [jta] | -0.153 / -0.240 |

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| dNf.i with aControl (mm) | [dNf0.i] | 28.380 | 109.793 |
| Reserve (dNf0.i-dFf.e)/2 (mm) | [cF0.i] | 0.043 | 0.260 |
| Tip clearance (mm) | [c0.i(aControl)] | 0.181 | 0.483 |

| | | | |
|---|---|---|---|
| Torsional angle Gear 2 with fixed Gear 1: | | | |
| Total torsional angle (°) | [j.tSys] | 0.2101 / 0.1138 | (0°12'36"/ 0°6'50") |
| Torsional angle Gear 1 with fixed Gear 2: | | | |
| Total torsional angle (°) | [j.tSys] | 0.7878 / 0.4269 | (0°47'16"/ 0°25'37") |

# 11 Gear tolerances

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| According to ISO 1328-1:2013, ISO 1328-2:1997 | | | |
| Accuracy grade | [Q] | A6 | A7 |
| Single pitch tolerance (µm) | [fpT] | ±8.50 | ±12.00 |
| Base pitch tolerance (µm) | [fpbT] | ±7.99 | ±11.28 |
| Sector pitch tolerance (µm) | [Fpz/8T] | ±19.00 | ±27.00 |
| Cumulative pitch tolerance, total (µm) | [FpT] | 24.00 | 40.00 |
| Adjacent pitch difference tolerance (µm) | [fuT] | 12.00 | 17.00 |
| Profile form tolerance (µm) | [ffαT] | 9.00 | 13.00 |
| Profile slope tolerance (µm) | [fHαT] | ±7.00 | ±10.00 |
| Profile tolerance, total (µm) | [FαT] | 11.00 | 16.00 |
| Helix form tolerance (µm) | [ffβT] | 11.00 | 16.00 |
| Helix slope tolerance (µm) | [fHβT] | ±9.50 | ±14.00 |
| Helix tolerance, total (µm) | [FβT] | 14.00 | 21.00 |
| Runout tolerance (µm) | [FrT] | 21.00 | 36.00 |

| | | | |
|---|---|---|---|
| Single flank composite tolerance, total (µm) | [FisT] | 32.00 | 51.00 |
| Single flank composite tolerance, tooth-to-tooth (µm) | [fisT] | 8.50 | 12.00 |
| Radial composite tolerance, total (µm) | [Fi"] | 26.00 | 43.00 |
| Radial composite tolerance, tooth-to-tooth (µm) | [fi"] | 9.50 | 13.00 |

$F_{idT}$ ($F_i$"), $f_{idT}$ ($f_i$") according to ISO 1328-2:1997 calculated with the geometric mean values for $m_n$ and d.

According to ISO 1328-2:2020

| | | | |
|---|---|---|---|
| Accuracy grade | [R] | R39 | R41 |
| Radial composite tolerance, total (µm) | [FidT] | 28.00 | 43.00 |
| Radial composite tolerance, tooth-to-tooth (µm) | [fidT] | 16.00 | 18.00 |

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99

| | | |
|---|---|---|
| Total helix deviation for calculation (µm) | [Fβ] | 21.00 |
| Distance between bearings for calculation (mm) | [LG] | 100.00 |
| Maximum value for axis deviation error (µm) | [fΣβ] | ± 21.88 |
| Maximum value for axis inclination error (µm) | [fΣδ] | ± 43.75 |

## 12 Modifications and determination of the tooth form

### 12.1 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

## 13 Supplementary data

| | | |
|---|---|---|
| Maximal possible centre distance ($\varepsilon_\alpha$ = 1.0) | [aMAX] | 73.583 |

### 13.1 Masses, stiffnesses and moments of inertia

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Mass (kg) | [m] | 0.268 | 1.626 |
| Total mass (kg) | [Σm] | 1.895 | |
| Moment of inertia for system, relative to the input: | | | |
| calculation without consideration of the exact tooth shape | | | |
| Gears individually ((da+df)/2...di) (kg*m²) | [J] | 3.896e-05 | 0.003992 |
| System (da+df)/2...di (kg*m²) | [J] | 0.0003229 | |

### 13.2 Wear, power loss, sound pressure level

| | | |
|---|---|---|
| Torsional stiffness at driving gear with fixed driven gear: | | |
| Torsional stiffness (MNm/rad) | [cr] | 0.138 |
| Torsion when subjected to nominal torque (°) | [δcr] | 0.029 |
| Sound pressure level based on Masuda, without PPTE/δs | [dB(A)] | 61.131 |

### 13.3 Power loss

| | | |
|---|---|---|
| Average coefficient of friction according to Niemann | [$\mu_m$] | 0.114 |
| Wear sliding coefficient by Niemann | [ζw] | 0.912 |
| Loss factor | [HV] | 0.214 |
| Gear power loss (kW) | [PVZ] | 0.211 |
| Meshing efficiency (%) | [ηz] | 97.569 |

## 14 Service life, damage

| | | |
|---|---|---|
| Required safety for tooth root | [$S_{Fmin}$] | 1.4000 |
| Required safety for tooth flank | [$S_{Hmin}$] | 1.0000 |
| Required service life | [H] | 1000.0000 |

Service life (calculated with required safeties):

| | | |
|---|---|---|
| System service life (h) | [$H_{att}$] | 1024 |

| | | ------- Gear 1 | ---------- Gear 2 ------- |
|---|---|---|---|
| Tooth root service life (h) | [$H_{Fatt}$] | 1e+06 | 1e+06 |
| Tooth flank service life (h) | [$H_{Hatt}$] | 1e+06 | 1024 |

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

### 14.1 Damage

Damage relative to the required service life (H, 1000.0 h)

| $D_{F1}$ (%) | <TD CENTER>$D_{F2}$ (%) | <TD CENTER>$D_{H1}$ (%) | <TD CENTER>$D_{H2}$ (%) |
|---|---|---|---|
| 0.0000 | 0.0000 | 0.0000 | 97.6108 |

Damage relative to the system service life ($H_{att}$, 1024.5 h)

| $D_{F1}$ (%) | <TD CENTER>$D_{F2}$ (%) | <TD CENTER>$D_{H1}$ (%) | <TD CENTER>$D_{H2}$ (%) |
|---|---|---|---|
| 0.0000 | 0.0000 | 0.0000 | 100.0000 |

## 15 Reliability calculation

### 15.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

### 15.2 Factors

| | | |
|---|---|---|
| Reliability of material data for $\sigma_{Hlim}$ (%) | [$R\sigma_{Hlim}$] | 99.00 |
| Reliability of material data for $\sigma_{Flim}$ (%) | [$R\sigma_{Flim}$] | 99.00 |

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

| Gear | Type | fac | <TD CENTER>β | <TD CENTER>$t_0$ | <TD CENTER>T | <TD CENTER>R(H) |
|---|---|---|---|---|---|---|
| | | cycles/h | | cycles | cycles | % |
| 1 | Tooth root | 72000 | 1.700e+00 | 9.654e+29 | 1.484e+30 | 100.0000 |
| 1 | Tooth flank | 72000 | 1.300e+00 | 9.014e+29 | 4.295e+30 | 100.0000 |
| 2 | Tooth root | 19200 | 1.700e+00 | 9.654e+29 | 1.484e+30 | 100.0000 |
| 2 | Tooth flank | 19200 | 1.300e+00 | 1.773e+07 | 8.448e+07 | 99.3018 |

fac = Number of load cycles per hour
β = Weibull shape parameter
$t_0$ = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

### 15.3 Resulting reliabilities and service lifes

| | | |
|---|---|---|
| Required service life (h) | [$H_{min}$] | 1000.0000 |
| Reliability R, tooth roots subsystem (%) | [$R_{subF}$] | 100.0000 |
| Reliability R, tooth flanks subsystem (%) | [$R_{subH}$] | 99.3018 |
| Reliability R, gears subsystem (%) | [$R_{subG}$] | 99.3018 |
| | | |
| Required reliability (%) | [$R_{min}$] | 99.0000 |
| Service life H, tooth roots subsystem (h) | [$H_{subF}$] | > 1'000'000 |
| Service life H, tooth flanks subsystem (h) | [$H_{subH}$] | 1024.4771 |
| Service life H, gears subsystem (h) | [$H_{subG}$] | 1024.4771 |

## 16 Remarks

### 16.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

### 16.2 Calculations and factors

- Details of calculation method:
  $c_\gamma$ according to Method B
  $K_v$ according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, $f_{sh}$ according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors $Z_L$, $Z_V$, $Z_R$, $Z_W$, $Z_X$, $Y_{\delta relT}$, $Y_{RrelT}$ and $Y_X$. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

---

End of report (lines: 702)

---